

EC3730 Final Project

You are to investigate and implement the Libre Router, which runs LibreMesh that layered on top of OpenWrt. OpenWrt is an open source operating system for the routers. You and your group are to do the following:

- 1) Do background research of Libre Router and LibreMesh; identify the weakness and vulnerability. Please search literature and review projects that are based on that.
- 2) Try to implement Libre Router and LibreMesh on virtual machines; see if you can create a network with the virtual routers. The network should have at least four Libre Routers.
- 3) Implement LibreRouter on Raspberry Pi; discuss the implementation obstacles and evaluate the performance of the Libre Routers; each group will have two Raspberry Pi and both groups must work together at the end to form the network of four routers allowing users to connect and communicate with each other.

Deliverables: IEEE format conference paper (~5 to 8 pages) of all your findings and evaluations; submit all the code and configuration files in the Sakai dropbox, so a new router can be setup easily. Due date: Last of Final (June 17, 2026)